

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 19-2024: A 46-Year-Old Man with Arthritis and Rash

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PRESENTATION OF CASE

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CME



Dr. Christopher V. Radcliffe (Medicine): A 46-year-old man was evaluated in the rheumatology clinic of this hospital because of arthritis and rash.

One week before the current presentation, an episode of lightheadedness occurred when the patient stood up from a seated position. The episode was associated with diaphoresis and mild nausea; it lasted for a few minutes and resolved spontaneously. Four days later, another episode of lightheadedness occurred, and the patient sought evaluation in the emergency department of this hospital.

The temporal temperature was 36.3°C, the blood pressure 147/80 mm Hg, the pulse 109 beats per minute, and the oxygen saturation 100% while the patient was breathing ambient air. Swelling of the wrists and ankles was present. The blood levels of glucose and electrolytes and the results of kidney-function tests were normal. The aspartate aminotransferase and alanine aminotransferase levels were normal; the alkaline phosphatase level was 157 U per liter (reference range, 45 to 115). The complete blood count with differential count was normal, as was the urinalysis. The erythrocyte sedimentation rate was 104 mm per hour (reference range, 0 to 14), the C-reactive protein level 6.3 mg per liter (reference value, <8.0), and the D-dimer level 990 ng per milliliter (reference value, <500).

An electrocardiogram showed sinus tachycardia. Computed tomography (CT) of the chest, performed after the administration of intravenous contrast material timed to evaluate the pulmonary arteries, showed no pulmonary embolus or pulmonary nodules; there was no mediastinal or hilar lymphadenopathy. Intravenous crystalloid solution was administered, and the lightheadedness resolved. Plans were made for an evaluation of the joint swelling and elevated erythrocyte sedimentation rate in the rheumatology clinic affiliated with this hospital. The patient was discharged home.

Three days later, the patient was seen in the rheumatology clinic. Additional history was obtained. Four months before the current presentation, mild coronavirus disease 2019 (Covid-19) had developed after the patient had returned from a vacation in Central America and the Caribbean. Symptoms associated with Covid-19 resolved after 5 days. One week after the resolution of symptoms, and 14 weeks

before the current presentation, pain and swelling of the wrists had developed. There was morning stiffness in both wrists and difficulty with forming a fist; the stiffness abated with activity and hot showers. The patient had had a partial response after the administration of topical diclofenac, acetaminophen, and plant-based topical applications for joint pain. He also had noted a sensation of instability in the ankles occurring toward the end of the day. In addition to the joint symptoms, a rash involving the chest, back, and legs had developed; the palms and soles were spared. The rash was mildly pruritic at night and nonpainful. Alopecia, anorexia, and an estimated 8 kg of weight loss had also occurred. Two weeks before the current presentation, nasal crusting had developed, and the patient had noted the presence of scant blood when using facial tissue.

The patient's medical history was notable for asthma and eczema. Medications included topical triamcinolone and topical hydrocortisone. An unspecified allergy to diphenhydramine had been reported previously. The patient lived in Boston and worked in an office. He smoked a half pack of cigarettes and consumed two alcoholic beverages daily. There was no history of illicit drug use. He was sexually active with men; his most recent sexual activity, with a new partner, had occurred 5 months before the current presentation. His family history included psoriasis in his grandmother.

On examination, the temporal temperature was 36.5°C, the blood pressure 156/90 mm Hg, the pulse 105 beats per minute, and the oxygen saturation 100% while the patient was breathing ambient air. Multiple small areas of alopecia were scattered throughout the scalp, and there was crusting in the distal nares. The left commissure of the mouth had an ulcerative lesion; there was no evidence of mucosal lesions or cervical lymphadenopathy. Swelling, warmth, and mild tenderness of the wrists and ankles were present. Swelling of several metacarpophalangeal joints was also present. There was a hyperpigmented, macular rash with scaling that involved the trunk (Fig. 1), back, and legs, extending to the dorsal aspect of the feet. The heart rate was tachycardic but regular, without murmurs. The remainder of the examination was normal.

Diagnostic tests were performed.



Figure 1. Clinical Photograph.

A hyperpigmented, macular rash with scaling is shown on the trunk. The rash also involved the back and legs, extending to the dorsal aspect of the feet.

DIFFERENTIAL DIAGNOSIS

Dr. April M. Jorge: This 46-year-old man with a family history of psoriasis, recent travel to the Caribbean and Central America, and a new male sexual partner was evaluated in the rheumatology clinic because of rash and symmetric, polyarticular inflammatory arthritis affecting the wrists, metacarpophalangeal joints of the hands, and ankles that had developed shortly after Covid-19 and had lasted for 14 weeks. Joint swelling and an elevated erythrocyte sedimentation rate had been observed during an evaluation in the emergency department for postural lightheadedness, which had resolved after treatment with intravenous fluids. I suspect that the postural lightheadedness was caused by hypovolemia, perhaps due to an underlying chronic inflammatory illness. The family history of psoriasis may indicate a predisposition to autoimmune diseases.¹ However, it is important to also consider infections, cancers, and drug reactions that can mimic systemic inflammatory rheumatic diseases. Because the patient's rash and symmetric, polyarticular inflammatory arthritis developed at the same time, I will create a differential diagnosis of conditions that can explain both manifestations, with the understanding that the most likely diagnosis would also account for the additional symptoms of alopecia, nasal crusting, and oral ulceration and the finding of a highly elevated erythrocyte sedimentation rate.

**AUTOIMMUNE AND AUTOINFLAMMATORY
RHEUMATIC DISEASES**

Both chronic polyarticular inflammatory arthritis and rash can be caused by various forms of autoimmune arthritis (including psoriatic arthritis and rheumatoid arthritis), systemic connective-tissue disease (including systemic lupus erythematosus [SLE] and dermatomyositis), systemic vasculitis, or systemic inflammatory syndromes. The morphologic features of the rash can help in narrowing the differential diagnosis. This patient had a hyperpigmented, macular rash with scaling that involved the trunk, back, and legs, extending to the dorsal feet. The rash was not consistent with plaque psoriasis, given the absence of plaques. Therefore, psoriatic arthritis would be an unlikely diagnosis. Systemic vasculitis — such as antineutrophil cytoplasmic antibody-associated vasculitis, IgA vasculitis, Behçet's disease, or cryoglobulinemia — is not of major concern because the rash was not consistent with palpable purpura (a finding that would be expected with small-vessel vasculitis) and the rash was not acneiform or nodular (features that would be expected with medium-vessel vasculitis). The rash associated with rheumatoid arthritis is palpable purpura that typically occurs in the context of rheumatoid vasculitis. If the patient had cutaneous dermatomyositis, I would expect the rash to involve the face or scalp. In addition, I would expect one or more typical features, such as photodistributed involvement of the upper back (shawl sign) or upper chest (V-neck sign) or Gottron's papules overlying the dorsal knuckles. The rash was not evanescent or salmon-colored, which would be expected with Still's disease.

Systemic Lupus Erythematosus

SLE is an important consideration in this patient, given that he had polyarticular inflammatory arthritis, a hyperpigmented rash with scaling, oral and nasal ulceration, patchy alopecia, and an elevated erythrocyte sedimentation rate.² However, several aspects of this patient's presentation are not consistent with a diagnosis of SLE. Discoid lupus causes skin hyperpigmentation with scaling. However, it also causes plaques and scarring, and it nearly always involves the head and neck; these features were not observed in this patient. Subacute cutaneous lupus can involve the trunk and legs while sparing the face. However, subacute cutaneous lupus is char-

acterized by small annular plaques or papules, and it typically involves sun-exposed areas including the arms.³ In addition, the oral ulceration that occurs in patients with SLE typically involves the hard and soft palate, tongue, and lower lip and does not involve the commissure of the mouth.

SLE is classically associated with an elevated erythrocyte sedimentation rate and a normal C-reactive protein level, possibly due to upregulation of type 1 interferons that inhibit the induction of C-reactive protein in hepatocytes.⁴ However, an erythrocyte sedimentation rate of more than 100 mm per hour is more common in patients with bacterial infection than in those with systemic autoimmune rheumatic disease or cancer.⁵ The erythrocyte sedimentation rate and the C-reactive protein level can fall within a wide range in patients with rheumatic diseases and in those with nonrheumatic diseases, so these values do not help in narrowing the differential diagnosis in this case. Overall, SLE is unlikely to explain this patient's presentation.

Sarcoidosis

Sarcoidosis can cause polyarticular inflammatory arthritis, macular rash, and alopecia. The most common form of sarcoid arthropathy is acute oligoarticular inflammatory arthritis, most often involving the ankles (Löfgren's syndrome), but in rare cases, sarcoidosis can cause chronic polyarticular inflammatory arthritis that affects the hands and wrists.^{6,7} Sarcoidosis can cause a variety of cutaneous lesions, including erythema nodosum, infiltrative nodules, plaques, and lupus pernio, as well as a red-brown maculopapular rash.^{8,9} Alopecia can occur but in the context of scarring from nodular cutaneous lesions involving the scalp, which was not observed in this patient. In addition, there were no pulmonary nodules, reticular opacities, or enlarged hilar lymph nodes on chest CT.

VEXAS Syndrome

VEXAS (vacuoles, E1 enzyme, X-linked, autoinflammatory, somatic) syndrome is a recently identified disorder caused by an X-linked somatic mutation in *UBA1* (the gene that encodes ubiquitin-like modifier activating enzyme 1) that causes a hematologic and autoinflammatory syndrome in patients older than 40 years of age.¹⁰⁻¹² Patients with VEXAS syndrome can present with inflam-

matory arthritis and a variety of inflammatory rashes, among other organ manifestations. Although the clinical presentation can be heterogeneous, VEXAS syndrome is unlikely in this patient because of the absence of macrocytic anemia, a hallmark feature.

NONRHEUMATIC DISEASES

Chronic polyarticular inflammatory arthritis can also be caused by drug reactions, cancer, and infection. The serum sickness drug reaction can cause rash and inflammatory arthritis that lasts for several weeks, but this patient's only medications were topical glucocorticoids, and there was no reported use of oral supplements. The morphologic features of the rash can again help in narrowing the differential diagnosis. This patient's rash was not consistent with erythroderma, so lymphoma is unlikely to explain his arthritis. His rash was not typical of those associated with vasculitis, so hepatitis B virus or hepatitis C virus infection is also unlikely. Although his symptoms developed shortly after Covid-19, chronic inflammatory arthritis is not a recognized manifestation of Covid-19.¹³ However, several other infections are important to consider.

Tuberculosis

Tuberculosis reactive arthritis (Poncet's disease) is sterile inflammatory arthritis that is typically polyarticular and symmetric and usually affects small joints, like the arthritis seen in this patient.¹⁴ This patient had no respiratory symptoms, and his findings on CT of the chest were not consistent with pulmonary tuberculosis. However, patients with extrapulmonary tuberculosis can also have reactive arthritis and tubercloid rashes from delayed-type hypersensitivity reactions. The most common tubercloid rash is erythema induratum of Bazin, which can occur with tuberculosis reactive arthritis. However, erythema induratum of Bazin is characterized by subcutaneous nodules, which were not seen in this patient.¹⁵ Lichen scrofulosorum is a tubercloid rash that can involve the trunk and legs, but it typically consists of many tiny papules with scaling that can coalesce into plaques. There is a papulonecrotic tubercloid rash that consists of hyperpigmented macules with scaling, like those seen in this patient. However, the macules usually become necrotic and typically involve the face or hands.¹⁶ This patient's rash was

not consistent with any of these tubercloid rashes, and he did not have fever. Overall, tuberculosis is unlikely in this case.

Arthritogenic Alphaviruses

Arthritogenic alphaviruses can cause chronic polyarticular inflammatory arthritis that can mimic rheumatoid arthritis or SLE. This patient may have been exposed to Chikungunya virus during travel in the Caribbean or exposed to Mayaro virus during travel in South America.¹⁷ Both viruses can cause inflammatory arthritis that can persist for several months or longer. However, the macular rash associated with infection with these viruses is typically transient, not persistent. In addition, this patient did not have fever or the myalgias and bone pain that are characteristic of infection with these viruses, so arthritogenic alphavirus infection is unlikely.

Parvovirus B19

Infection with parvovirus B19 can also cause symmetric, polyarticular inflammatory arthritis affecting small and large joints, like the arthritis seen in this patient or in those with rheumatoid arthritis or SLE. The inflammatory arthritis typically occurs 1 to 3 weeks after an acute febrile illness and resolves within 6 weeks. A corresponding rash is often present at the same time and may be characterized by confluent erythema or by macules and papules on the trunk, arms, and legs in adults, as opposed to the classic "slapped cheek" appearance of erythema infectiosum seen in children.¹⁸ Although the inflammatory arthritis and rash in this patient are consistent with parvovirus B19 infection, the symptom duration of at least 14 weeks does not fit with the time course of that illness. Also, parvovirus B19 is not known to cause patchy alopecia.

Syphilis

Syphilis is known to mimic SLE and other systemic autoimmune rheumatic diseases. The incidence of syphilis is highest among men who have sex with men but has been rising in the United States among men regardless of the sex of their sexual partners and among women. From 2017 to 2021, there was a 74% increase in the overall number of cases of syphilis in the United States.¹⁹ Syphilis can cause oligoarticular or polyarticular inflammatory arthritis, and the second-

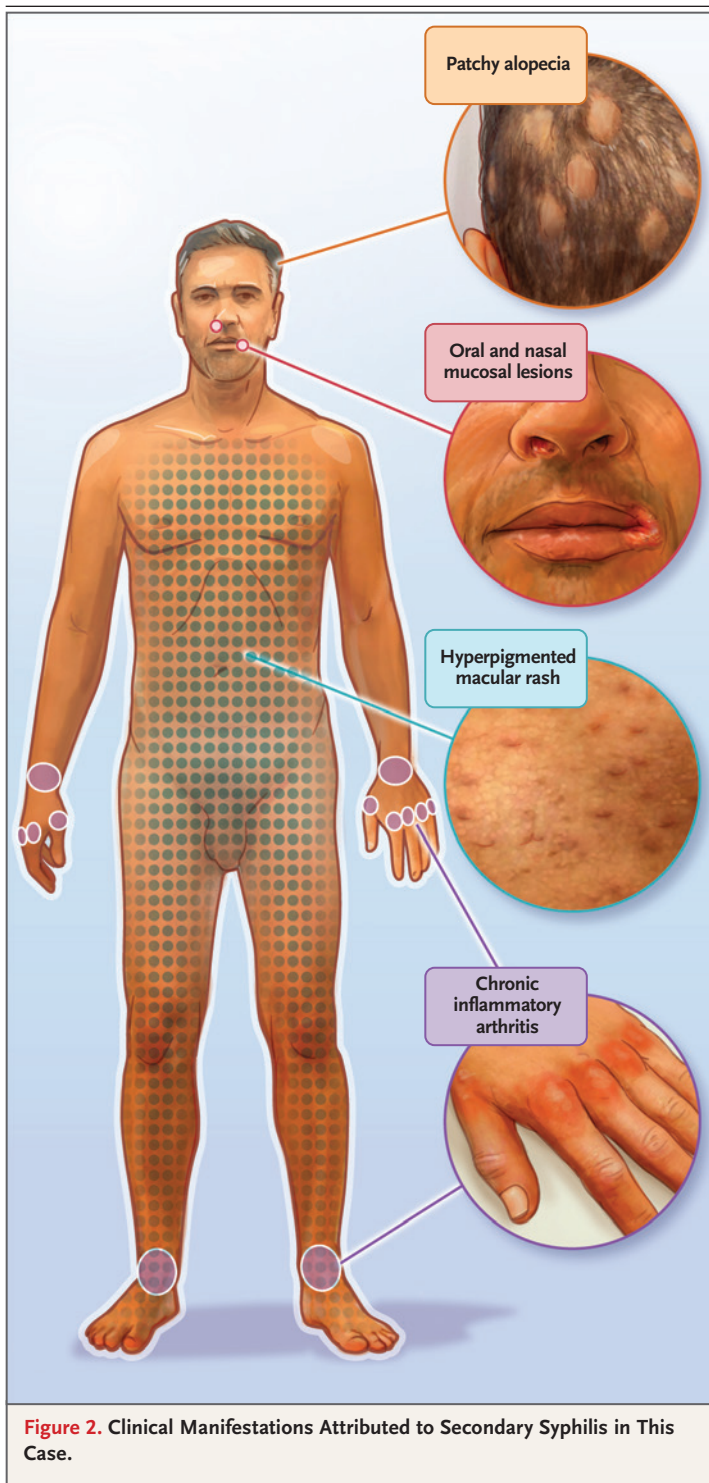


Figure 2. Clinical Manifestations Attributed to Secondary Syphilis in This Case.

any stage of syphilis can result in a macular or papular rash with scaling on the trunk, arms, and legs (Fig. 2).²⁰ Although the rash associated with secondary syphilis classically involves the

palms and soles, this is not always the case. A helpful diagnostic feature in this patient is the description of alopecia. Syphilis can cause reversible patchy alopecia with a classic “moth-eaten” appearance, with diffuse, small patches of hair loss.^{21,22} Syphilis could also explain the ulceration of the oral and nasal mucosa seen in this patient. The most common oral manifestation of secondary syphilis is mucous patches involving the tongue, lips, or palate that can look like aphthous ulcers. However, syphilis can also cause unilateral lesions at the oral commissure, a finding seen in this patient.²³ Sinonasal involvement is more typical of tertiary syphilis, but nasal ulceration is occasionally seen in patients with secondary syphilis.²⁴ Overall, secondary syphilis is the most likely diagnosis in this patient.

To establish this diagnosis, I would perform a treponema-specific antibody test and a rapid plasma reagin (RPR) test.

DR. APRIL M. JORGE'S DIAGNOSIS

Secondary syphilis.

DIAGNOSTIC TESTING

Dr. Sarah E. Turbett: The first diagnostic test performed in this case was a treponemal antibody test, which was reactive. This result was confirmed with an RPR test, which was reactive, with a titer of 1:256. In combination with the clinical presentation, these results are diagnostic of polyarthritis due to secondary syphilis.

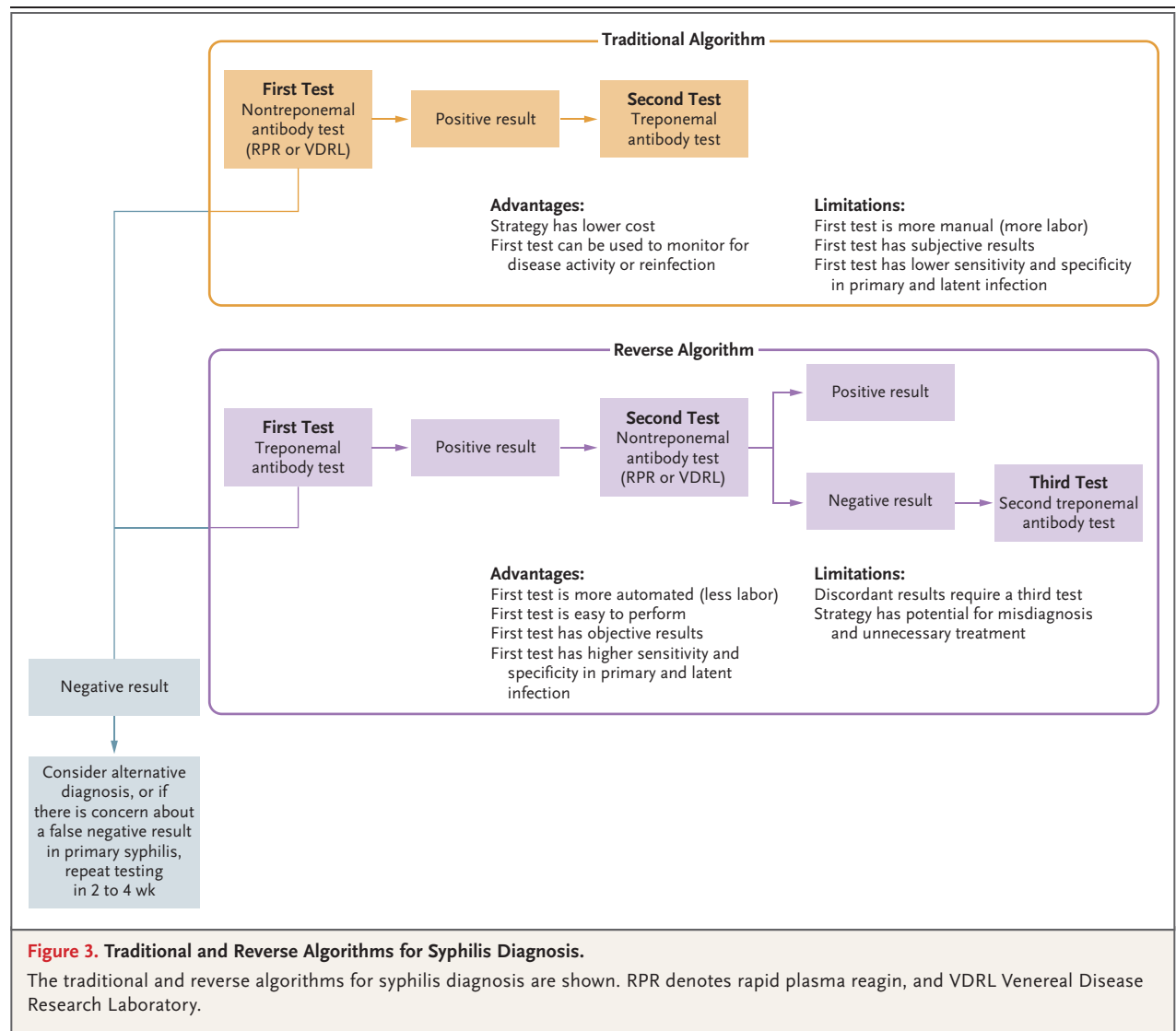
Syphilis is an infection caused by the spirochete *Treponema pallidum*, which is transmitted primarily through sexual contact.²⁵ The disease has three stages, each with a characteristic clinical syndrome. Primary infection has a hallmark appearance of a single or multiple painless ulcerative skin lesions, known as chancres, which can persist for weeks.^{25,26} Secondary infection has a more varied presentation, including articular manifestations, as were seen in this case.^{25,26} Tertiary infection is commonly latent, but clinical manifestations such as cardiac or neurologic involvement or gumma formation may occur.²⁵

The cornerstone of syphilis diagnosis is serologic testing, with both treponemal and nontreponemal antibody tests performed as part of a tiered algorithm to increase diagnostic specificity.²⁵ Nontreponemal antibody tests — the RPR

test and the Venereal Disease Research Laboratory test — target antibodies to tissue lipid proteins produced during infection, whereas treponemal tests detect antibodies to *T. pallidum*-specific antigens.^{25,27} Nontreponemal tests are rapid, relatively inexpensive, and easy to perform, and they have a reported sensitivity that approaches 100% in patients with secondary syphilis. Nontreponemal tests are also quantitative, and serial testing can show a change in titer that is useful in monitoring the treatment response and assessing for reinfection.^{25,27} Next-generation treponemal antibody tests are similarly relatively inexpensive and easy to perform. They have greater sensitivity and specificity than nontreponemal tests in patients

with early or latent disease. Unlike nontreponemal tests, treponemal tests remain positive even after successful treatment, which limits their usefulness in distinguishing between active and past infection.^{25,27}

The Centers for Disease Control and Prevention endorses two strategies for tiered testing: the traditional algorithm, in which a reactive nontreponemal antibody test is confirmed by a treponemal antibody test, and the reverse algorithm, in which a reactive treponemal antibody test is confirmed by a nontreponemal antibody test (Fig. 3).²⁵ For the reverse algorithm, when the results of the two tests are discordant, a second treponemal antibody test with orthogonal prop-



erties is performed for adjudication. Both algorithms have good clinical performance in patients with secondary or tertiary disease, and there are advantages and limitations to both. Advantages of the traditional algorithm include its lower cost and the ability of the first test to distinguish between active and past infection.²⁵ Notable limitations include a more manual first test that has subjective results and lower sensitivity and specificity in primary and latent disease.²⁵ Advantages of the reverse algorithm include a more automated, easy-to-perform first test that has objective results and higher sensitivity and specificity in primary and latent disease.²⁵ The main limitations of the reverse algorithm include the need for a third test if there are discordant results and the potential for misdiagnosis and unnecessary treatment in populations with a high likelihood of previous syphilis.²⁵

LABORATORY DIAGNOSIS

Secondary syphilis.

DISCUSSION OF MANAGEMENT

Dr. Kevin L. Ard: The first step in the management of syphilis is to determine the stage of infection and, regardless of stage, whether neurosyphilis is present. This requires a detailed assessment of symptoms, including inquiring about ocular and otic symptoms, and a careful examination. This patient did not have any signs or symptoms of neurosyphilis. Parenteral penicillin is the preferred therapy for all forms of syphilis.²⁶ The recommended treatment for secondary syphilis — which was seen in this patient — would be a single dose of long-acting, intramuscular benzathine penicillin G. It is also prudent to perform a repeat RPR test on the day of treatment, even if only a few days have elapsed since diagnosis. The rationale for performing repeat testing is that RPR titers can change rapidly, so the titer ob-

tained at the time of treatment will provide the best baseline against which later titers can be compared.²⁸ After treatment of primary or secondary syphilis, the RPR titer is expected to decrease by a factor of at least 4 within 12 months.²⁶

Beyond staging and antimicrobial therapy, the management of syphilis also consists of several additional steps. Partners who have had sexual contact within the past 90 days with a person with secondary syphilis should be offered testing and empirical treatment; disease intervention specialists can assist with partner notification, should that be desired by the patient.²⁶ Patients with syphilis should also be assessed for common associated infections, including human immunodeficiency virus (HIV) infection, chlamydia, and gonorrhea. For those who do not have HIV infection, I would consider HIV preexposure prophylaxis (PrEP), given that syphilis is associated with incident HIV infection.²⁹⁻³¹ Finally, when syphilis occurs in men who have sex with men, the patients may benefit from postexposure prophylaxis with doxycycline after sexual encounters to prevent syphilis, gonorrhea, and chlamydia going forward.³²

After this patient received penicillin for secondary syphilis, the pain and swelling in the ankles, left wrist, and hands resolved, as did the rash, nasal crusting, and alopecia. The symptoms in the right wrist abated but did not fully resolve. Testing for HIV infection, gonorrhea, and chlamydia was negative, and he began PrEP with emtricitabine–tenofovir disoproxil fumarate. By 7 months after treatment for syphilis, his RPR titer had decreased to 1:8.

FINAL DIAGNOSIS

Secondary syphilis.

This case was presented at the Medicine Case Conference. Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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